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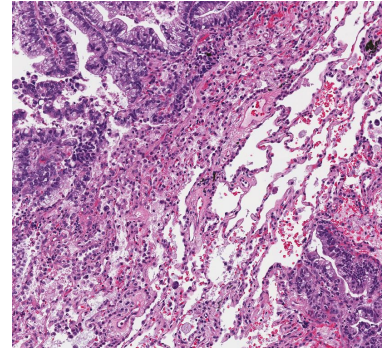
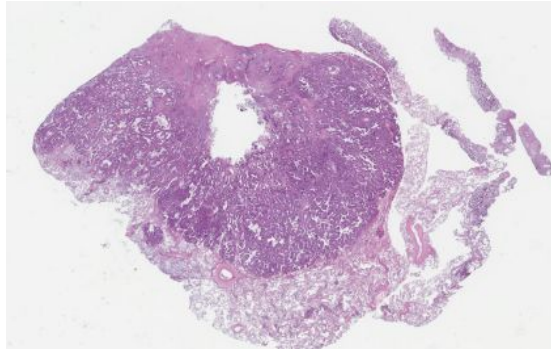
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Segmentation and analysis of cell distribution in cancer tissue

FELLER Matis

- Huge progress in computer vision
- Image analysis : crucial in the medical field (large number, complexity)
- Particularly in oncology
- Analysis of biopsies is expensive, laborious and error-prone procedures.



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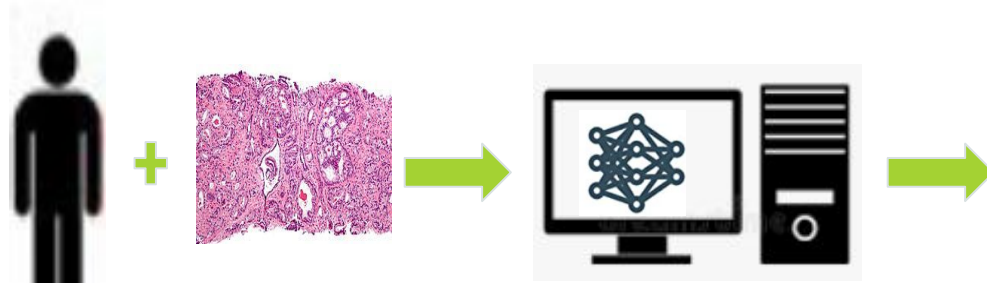


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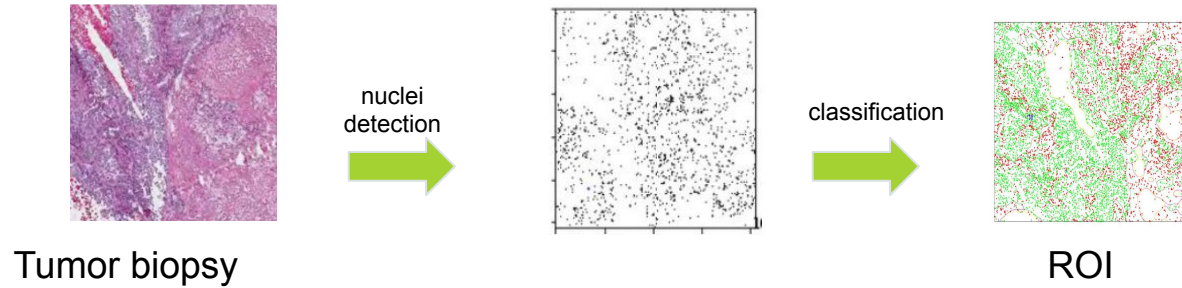
I. Objectives



Analysis of the tumor
microenvironment

- Medically : Allows to orient treatments & therapeutic for a new patient compared to the base we have. And provide precise observation points to watch.
- Cancer comprehension : Comparison methodology which will take into account the availability of cells and their interactions. Suggest new and more decisive features

- Digitize visual information (segmentation + classification)



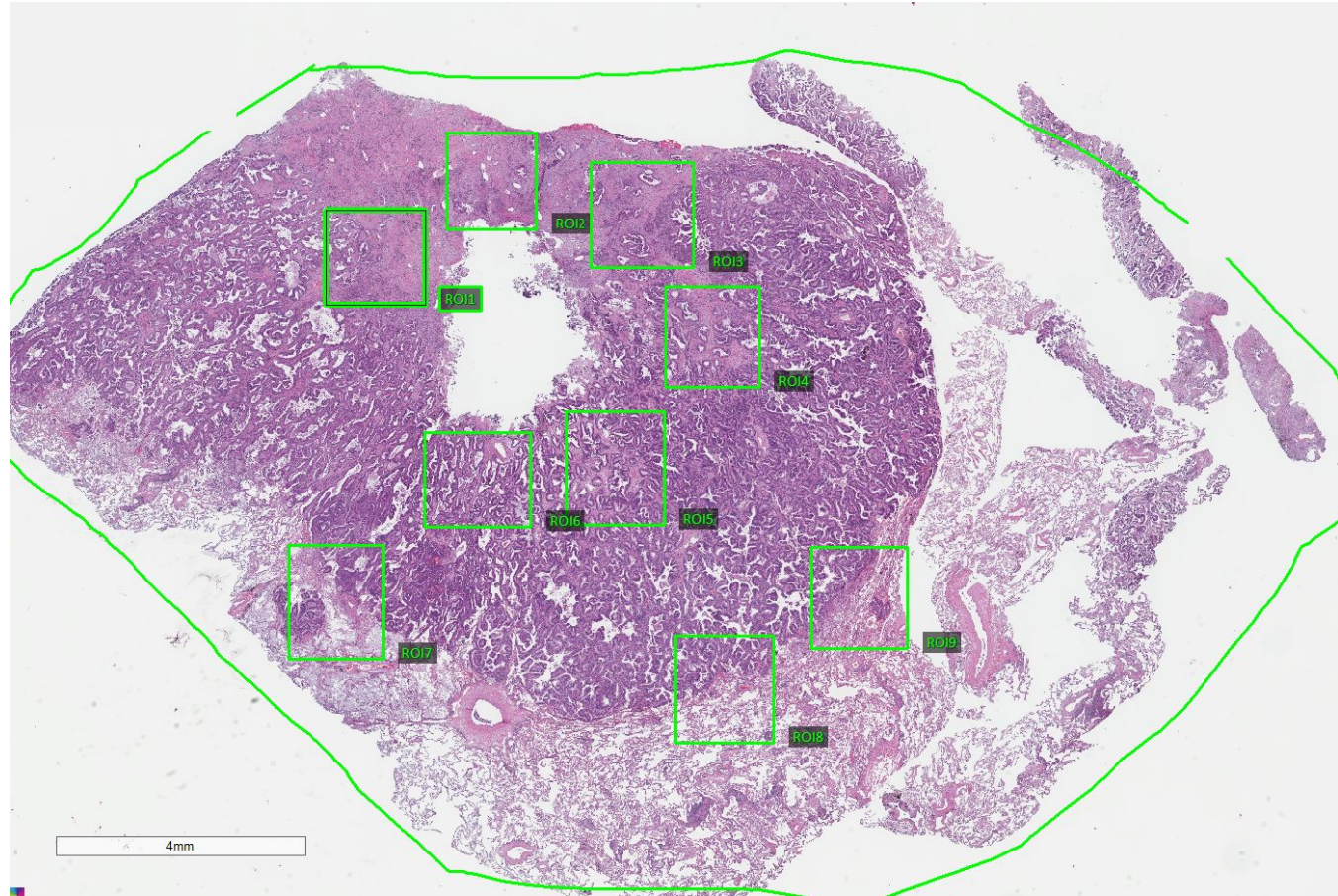
- **Create an ROI analysis tool capable of classifying patients**



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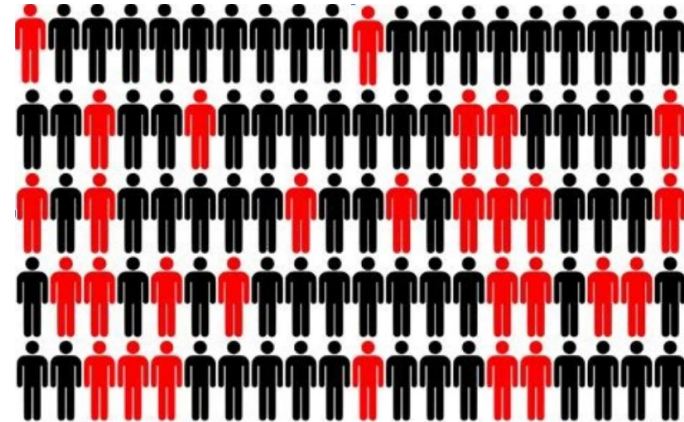
II. Labeling, Segmentation & Database

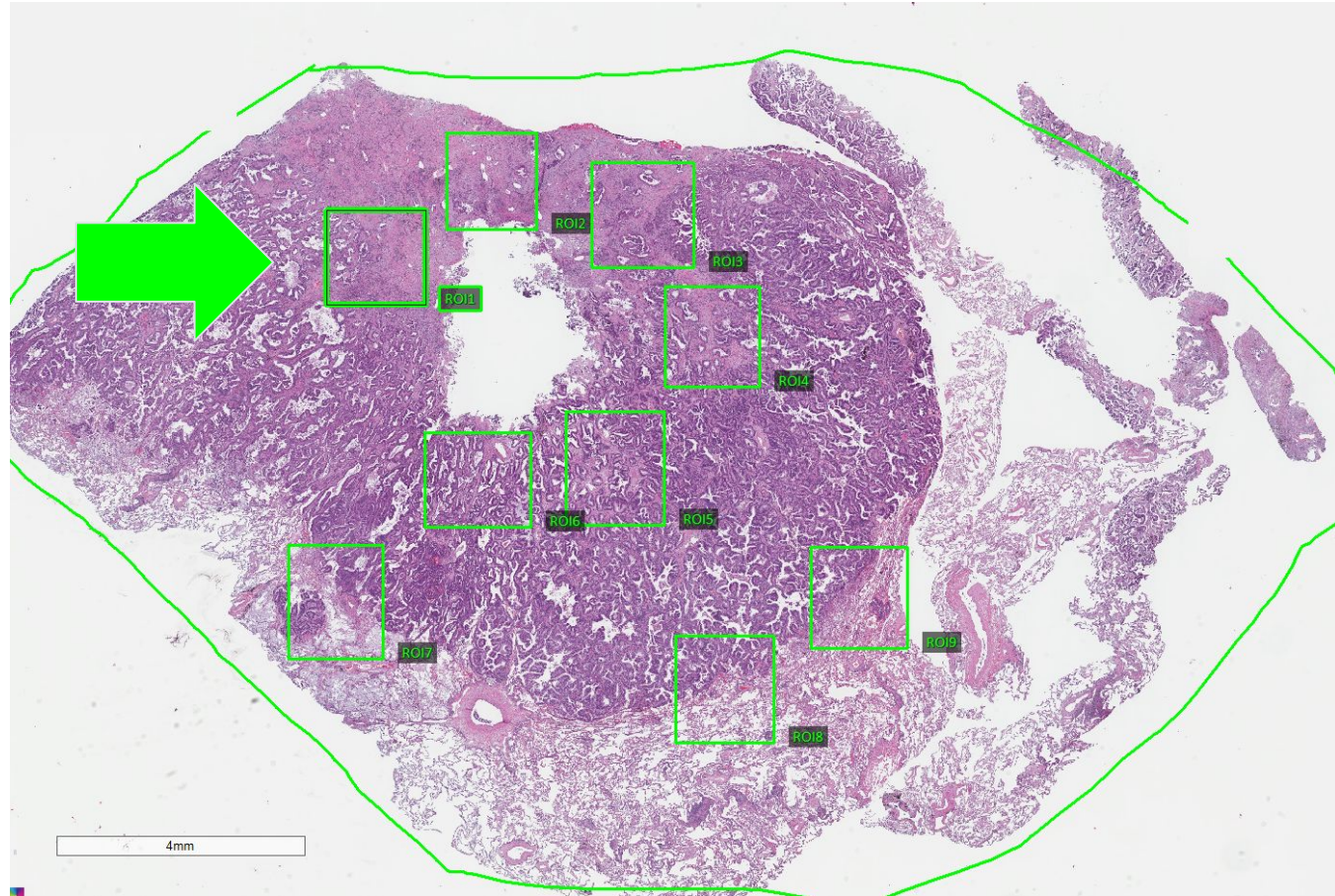


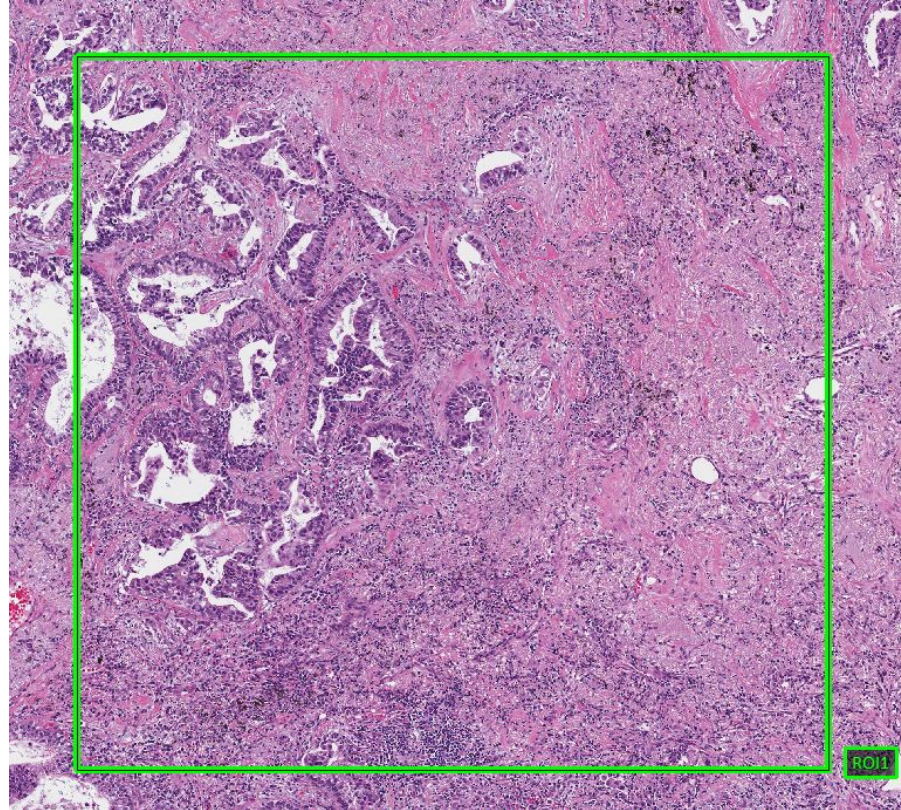
- 140 patients at stage I (I-A & I-B)
- infiltrative adenocarcinoma of the lung from The Cancer Genome Atlas (TCGA)
- 140 -> 115 usable patients for a reliable survival analysis

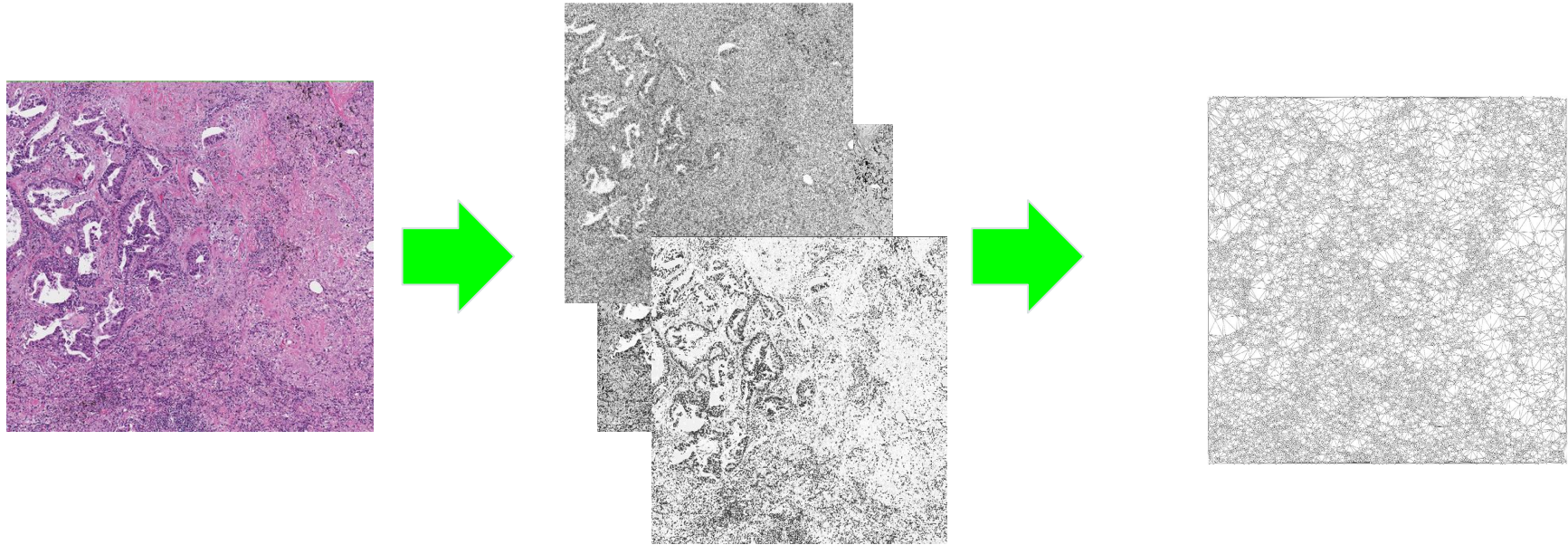


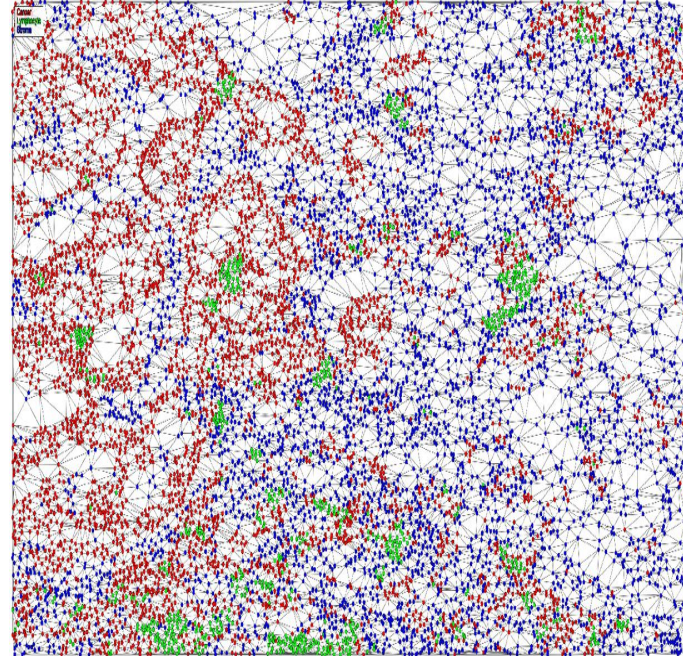
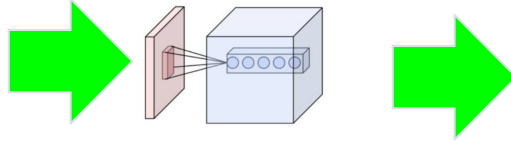
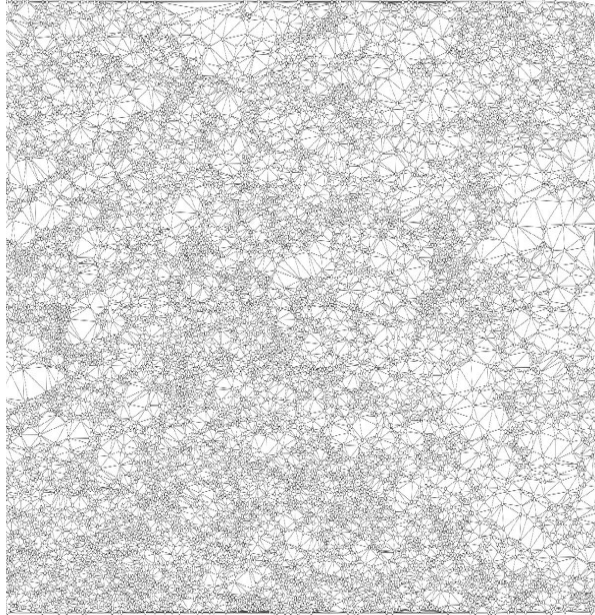
# id patient	Vivant / Mort	Days to last follow up	File
1	T	50	patient1.svs
2	F	150	patient2.svs
3	T	45	patient3.svs
...	...	...	...



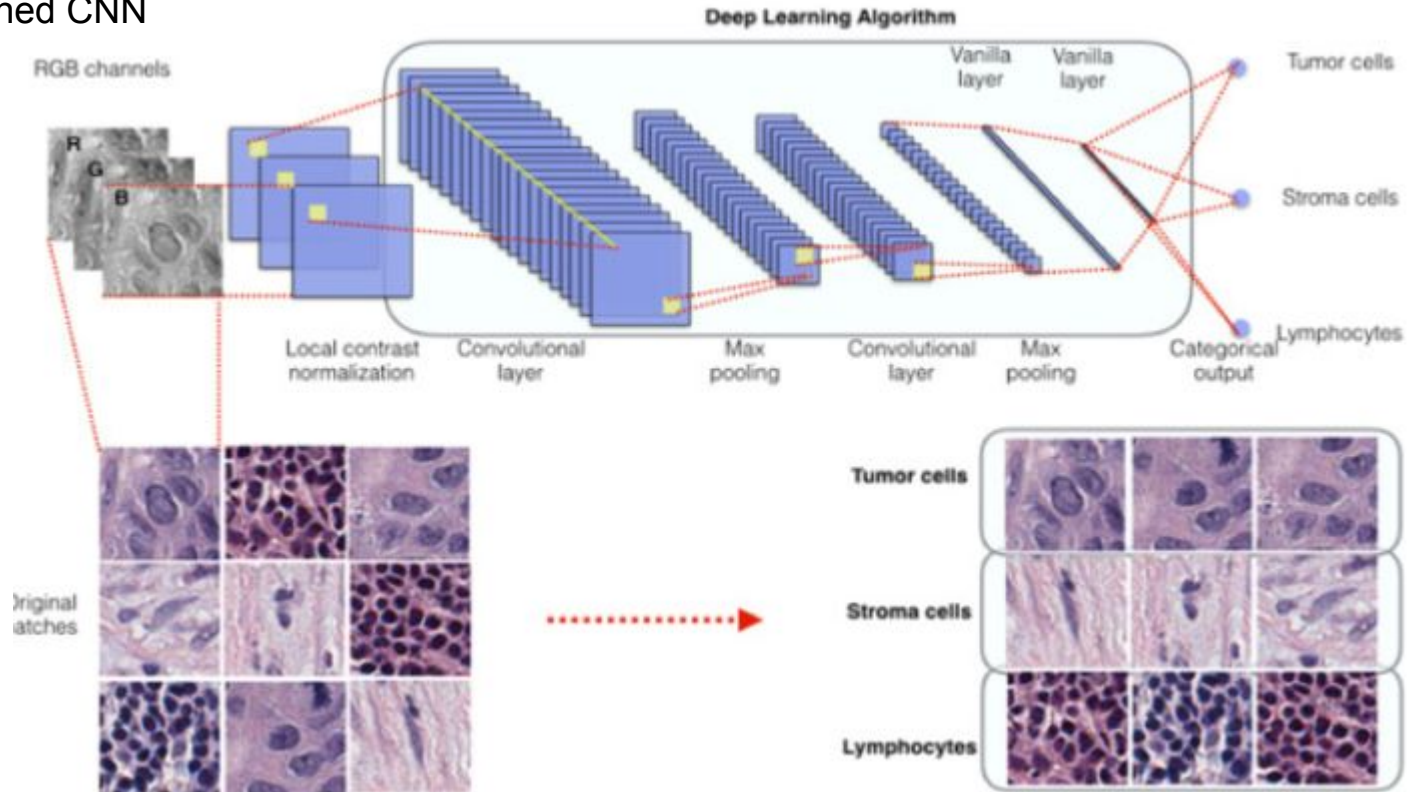








[1] Pretrained CNN





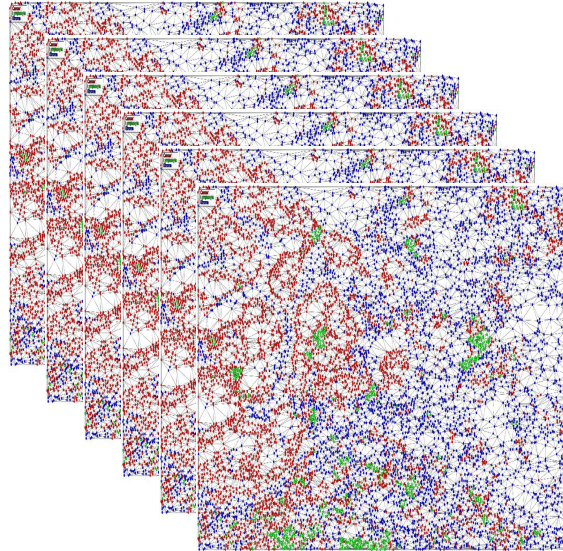
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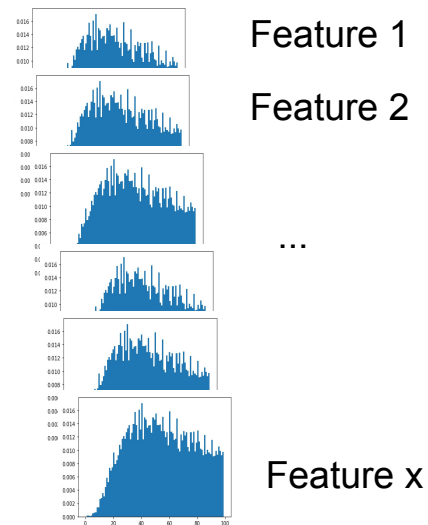
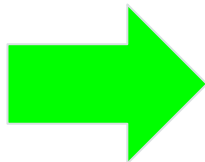
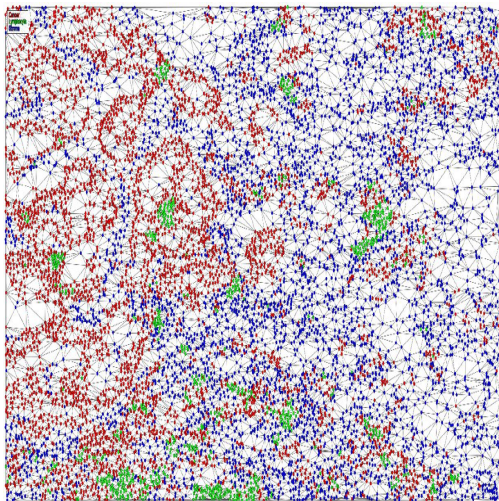
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III. Analysis of the tumor micro-environment using histograms

- 1) We start from our ROIs (1 patients = many ROIs)

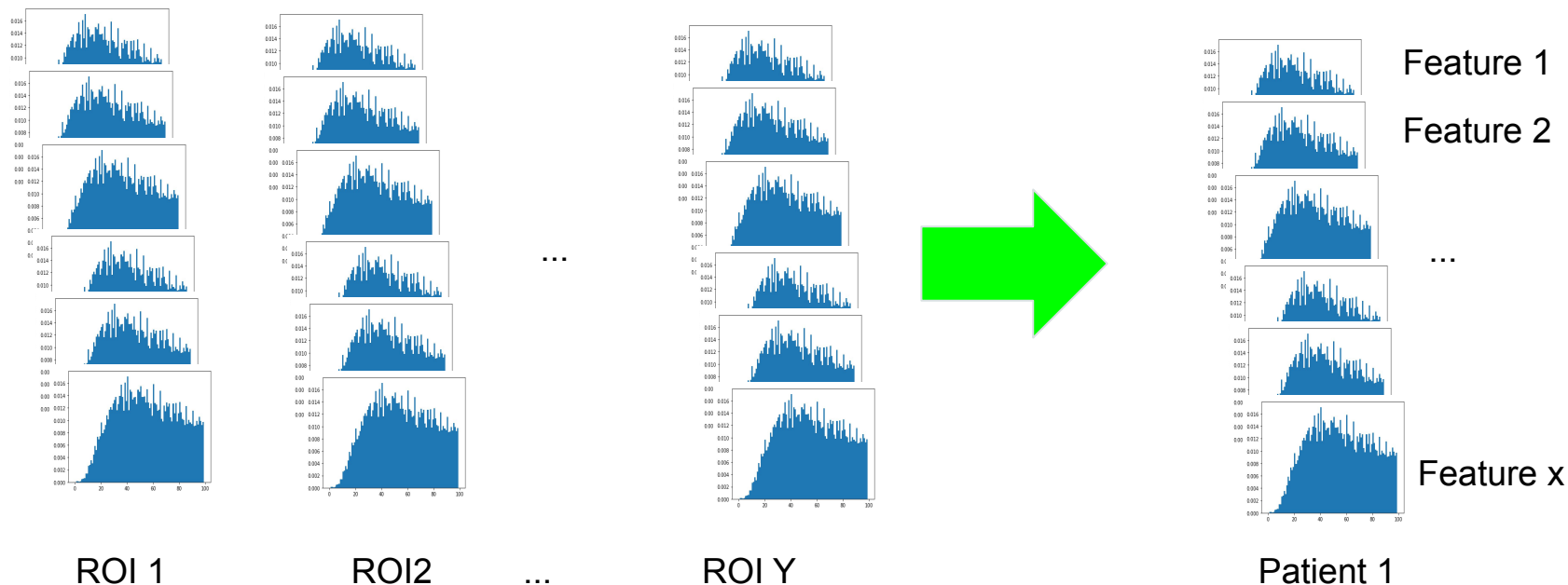


2) We extract the distribution of several features characterizing our micro-environment in the form of a histograms

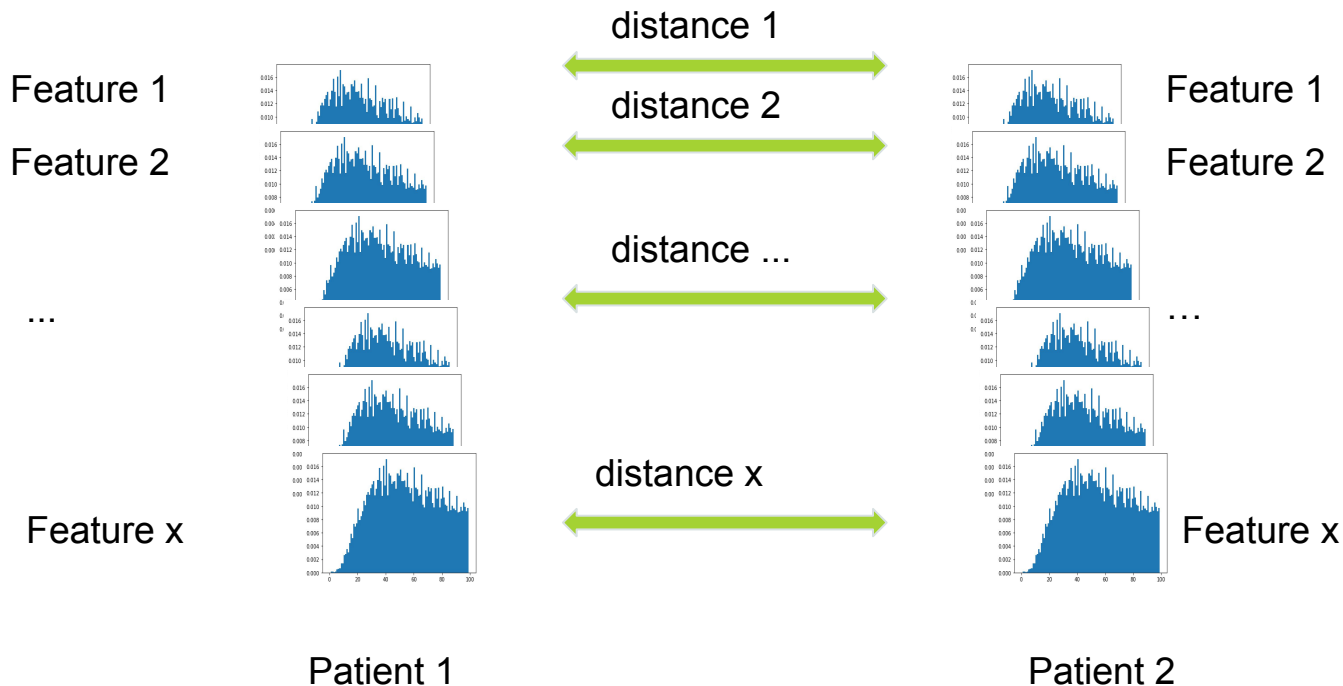


Exemple of feature : Lymphocyte-Cancer distance repartition

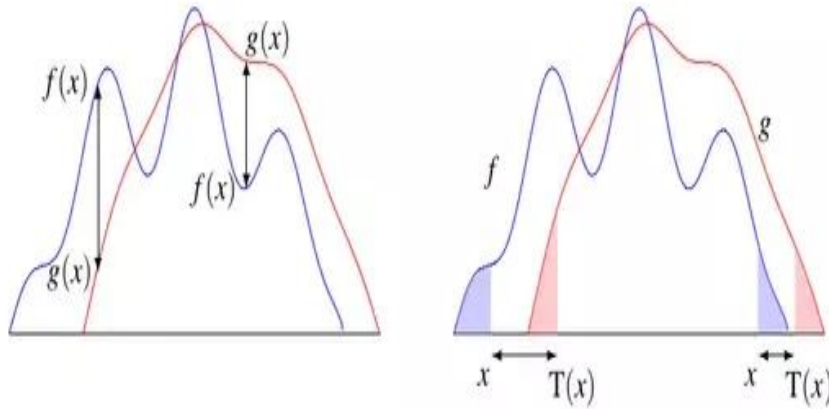
3) The ROI distributions of the same patient are aggregated



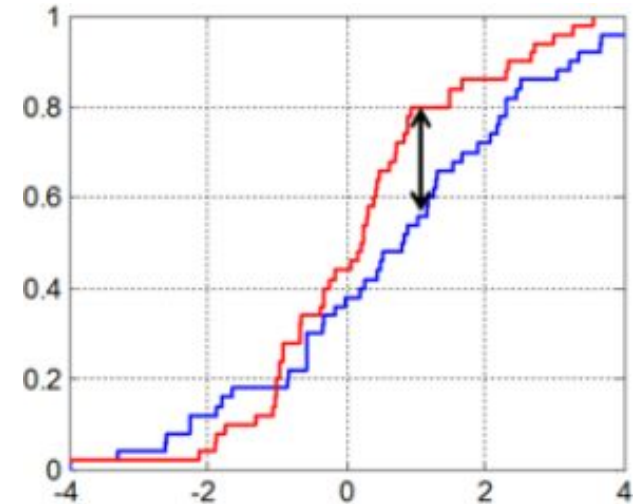
3) We compute the distance between each patients for each feature



1. Distance de Wasserstein



2. KS distance



(3. KbL divergence)

3) So basically, we have a distance matrix per feature

$$\text{Distance Feature 1} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & \dots(\text{patients}) \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ \dots \end{matrix} & \begin{pmatrix} D_{11} & D_{12} & D_{13} & D_{14} \\ D_{21} & D_{22} & D_{23} & D_{24} \\ D_{31} & D_{32} & D_{33} & D_{34} \\ D_{41} & D_{42} & D_{43} & D_{44} \\ \dots & & & \dots \end{pmatrix} \end{matrix}$$

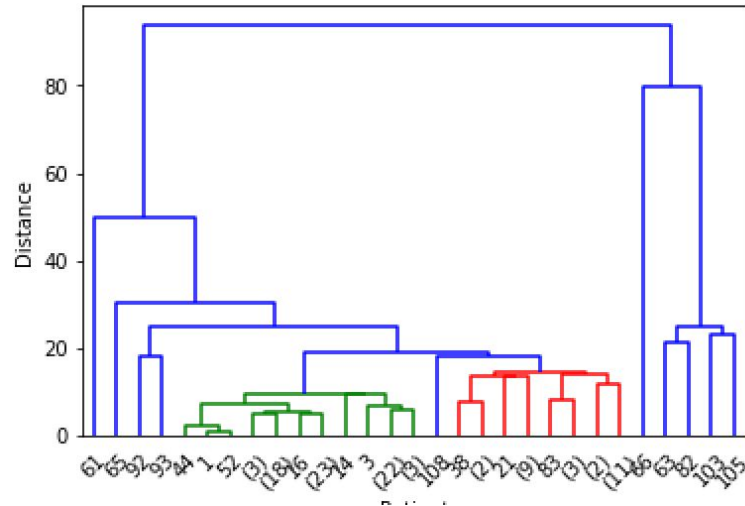
(patients)

$$\text{Distance Feature X} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & \dots \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ \dots \end{matrix} & \begin{pmatrix} D_{11} & D_{12} & D_{13} & D_{14} \\ D_{21} & D_{22} & D_{23} & D_{24} \\ D_{31} & D_{32} & D_{33} & D_{34} \\ D_{41} & D_{42} & D_{43} & D_{44} \\ \dots & & & \dots \end{pmatrix} \end{matrix}$$

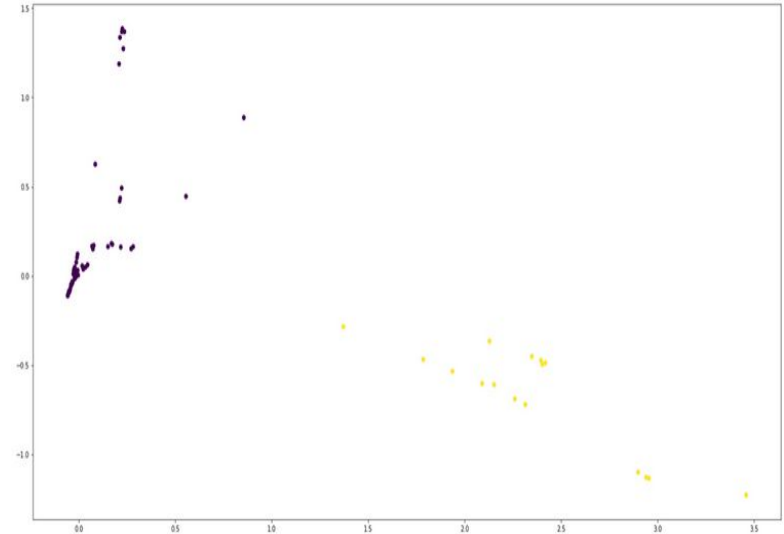
$$\text{Mean ponderate distance} = \frac{w_1 * \text{Distance Feature 1} + w_2 * \text{Distance Feature 2} + \dots + w_X * \text{Distance Feature X}}{X}$$

5) We perform a clustering based on this matrix to separate 2 groups

Dendrogramme

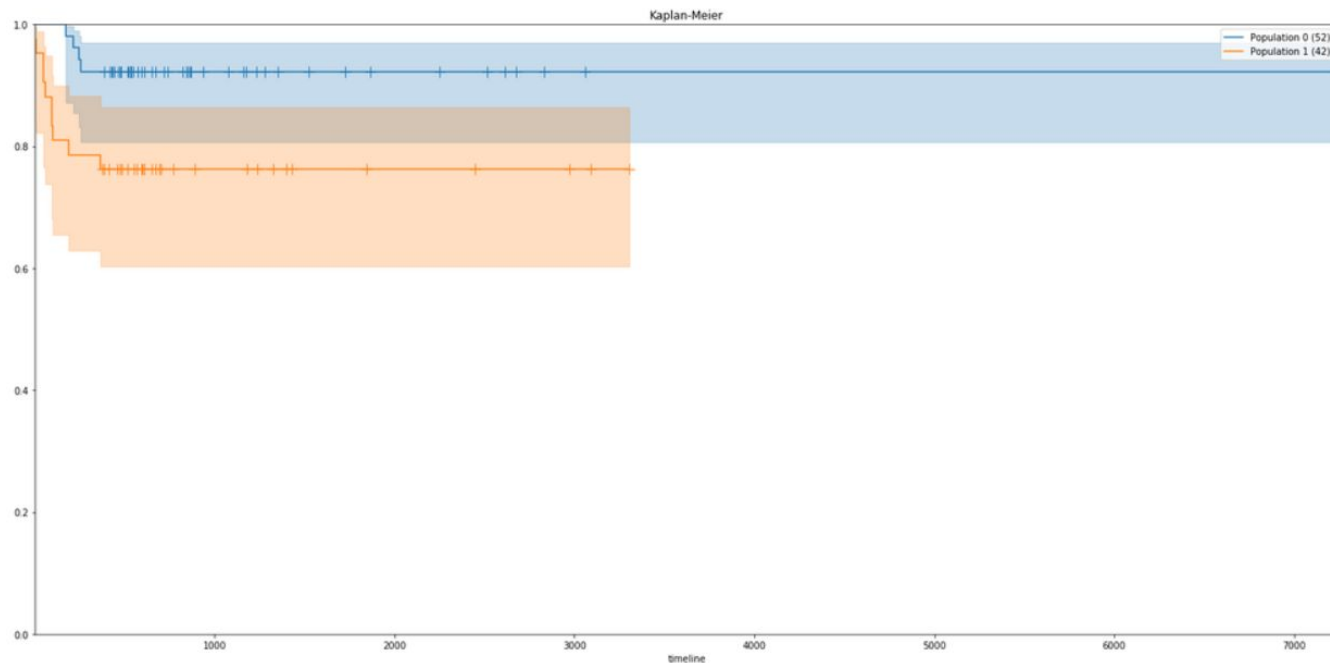


CAH



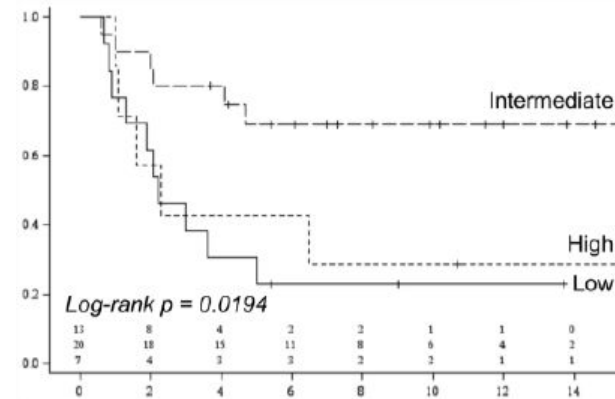
Spectral clustering

6) We evaluate the relevance of our clustering using survival curves with the Kaplan-Meier estimator



7) We check the reliability of our results by observing if they are significant using the p-value of the log rank test

$$\chi^2 = \sum_{i,j} \frac{(n_{ij} - n_{ij}^*)^2}{n_{ij}^*}$$



Logrank test = estimator which verifies reliably if the curves are statistically

Features extraction



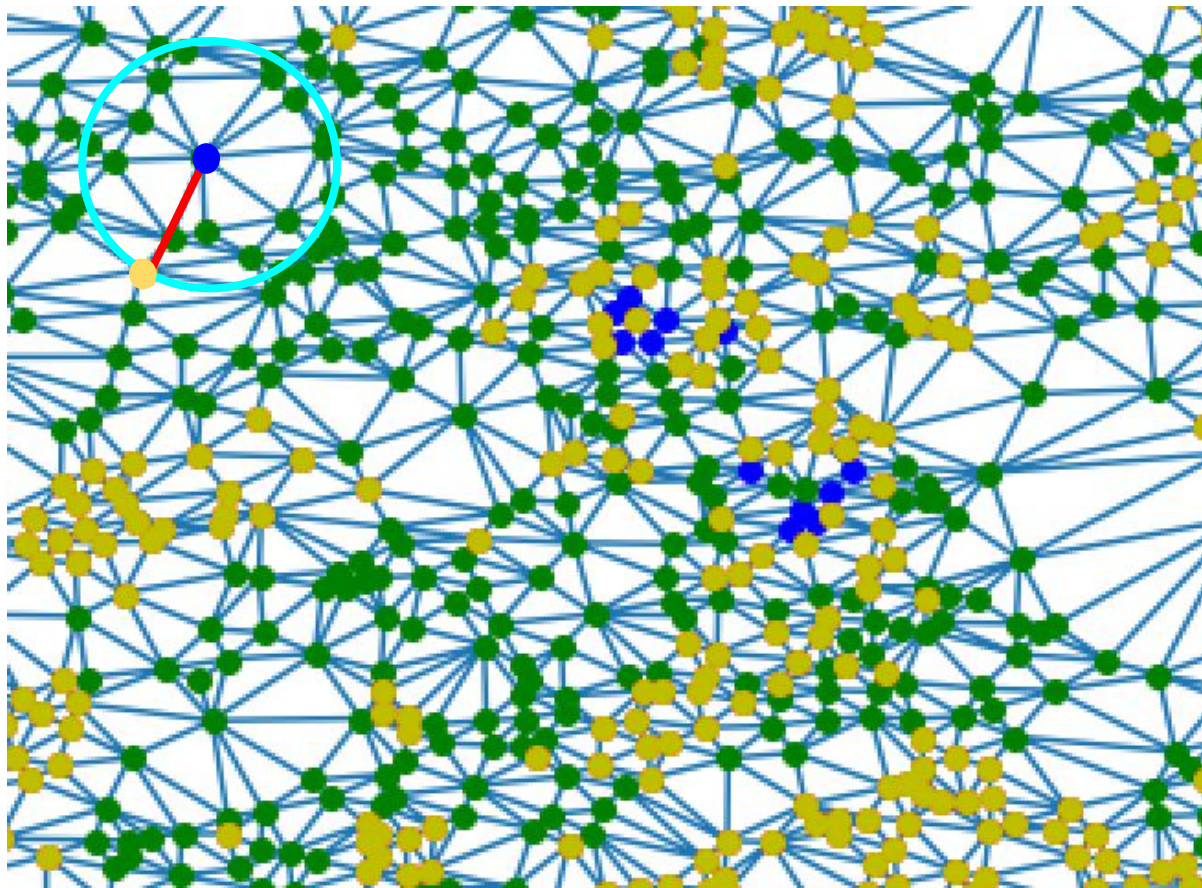
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III.B. Features extraction

1) Distances

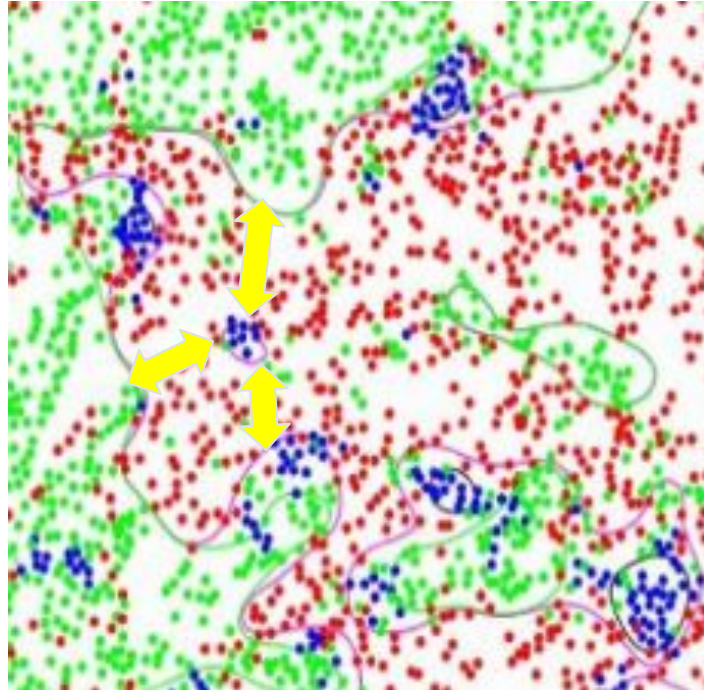
26

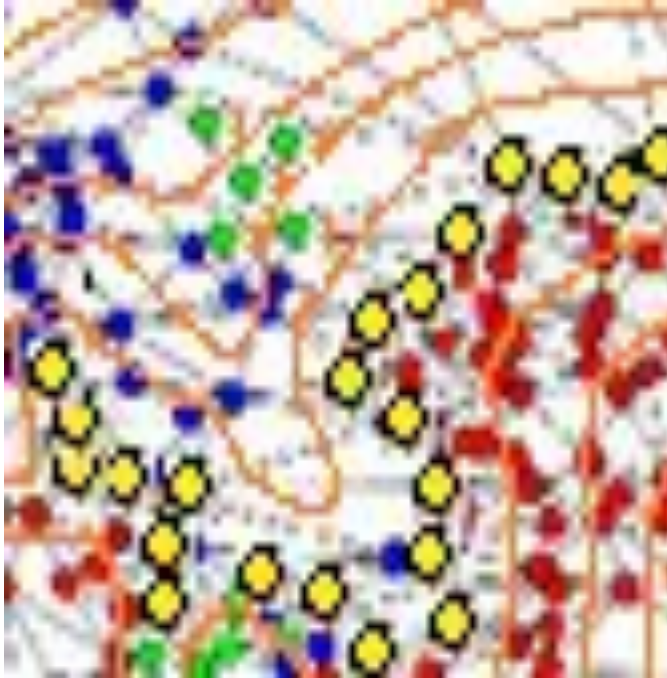
- a) Minimum distance
between 2 types of
cells



1) Distances

b) Distances to the tumor border





[5,10] μm = 1 Lympho

[0-5] μm = 0 Lympho

[-5,0] μm = 2 Lympho

[-10,-5] μm = 2 Lympho

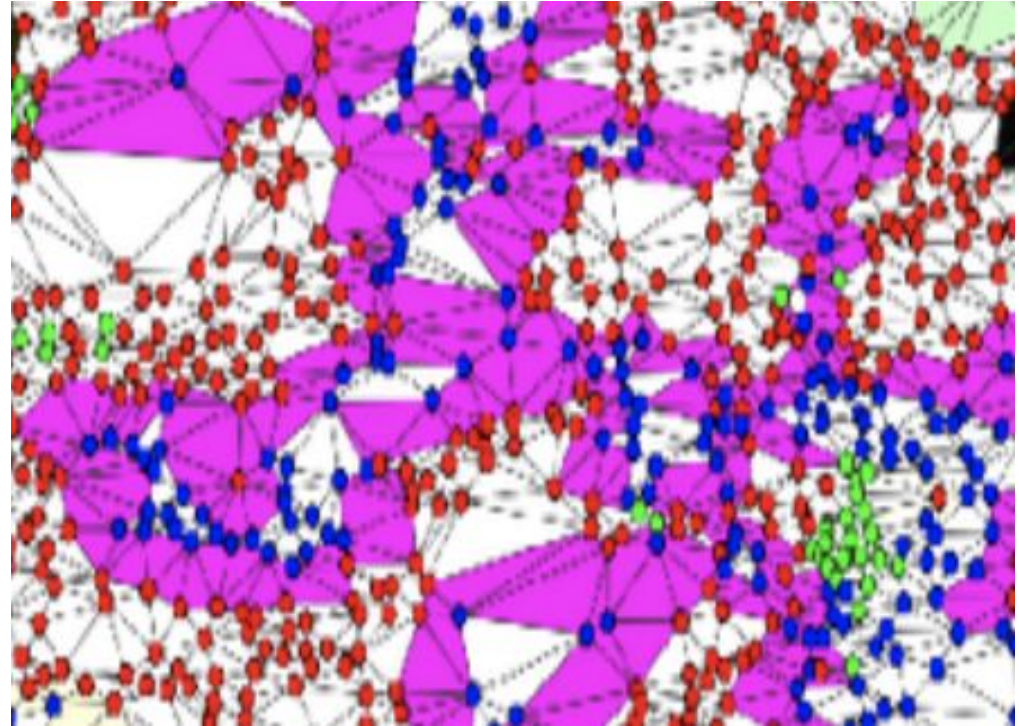
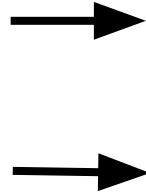
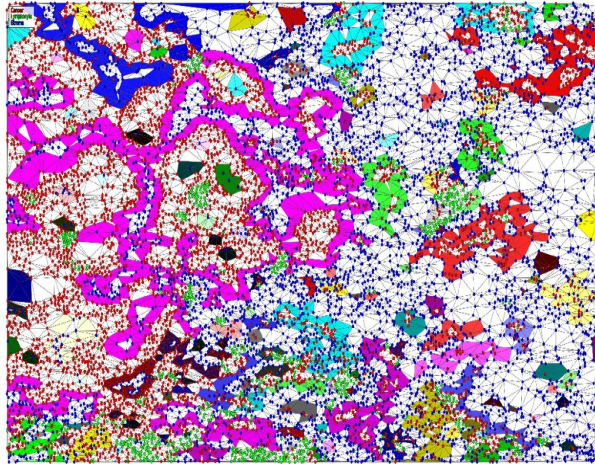
[-10,-5] μm = 0 Lympho

Normalized by the number
of cells for each bands

III.B. Features extraction

3) Areas at the band of the tumor interface

29

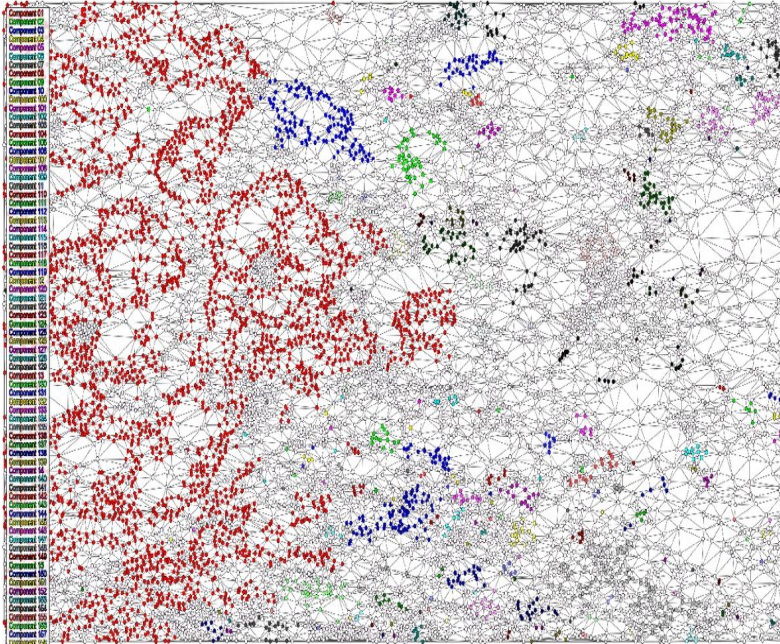


III.B. Features extraction

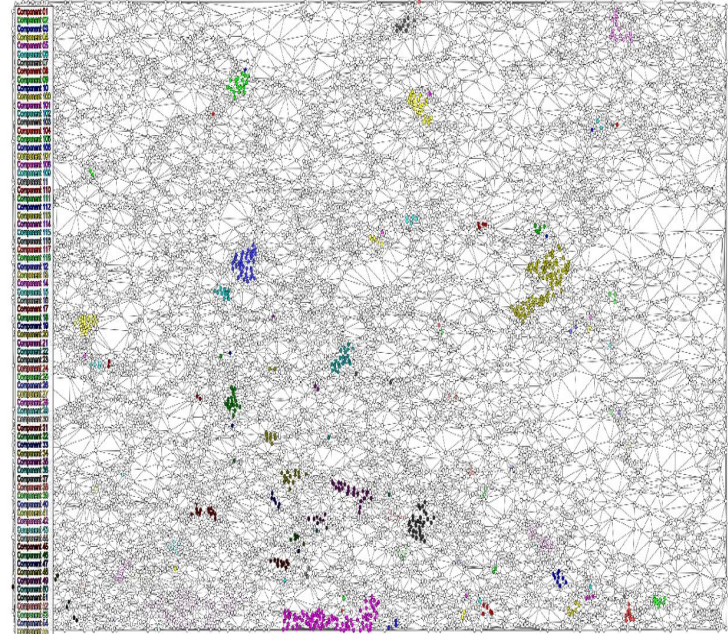
30

4) Size of clusters of cells of the same type

Cancers blobs



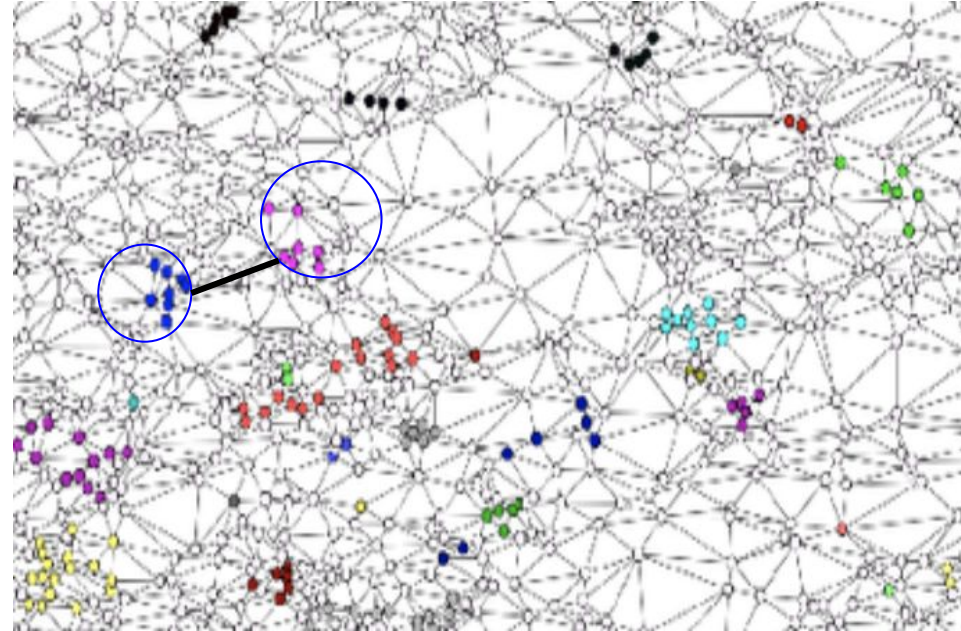
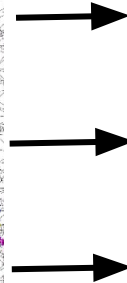
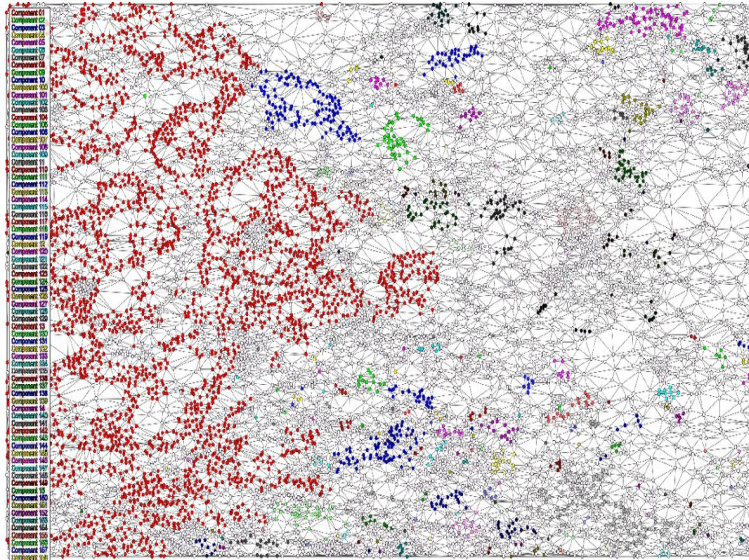
Lymphocytes blobs



III.B. Features extraction

5) Distance inter “blobs”

31



Results

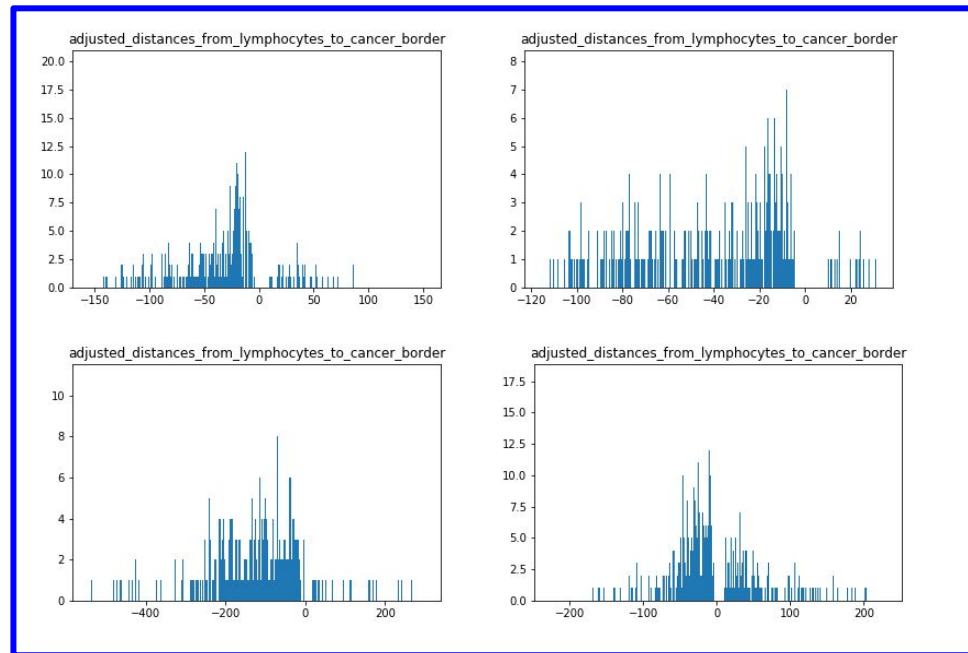


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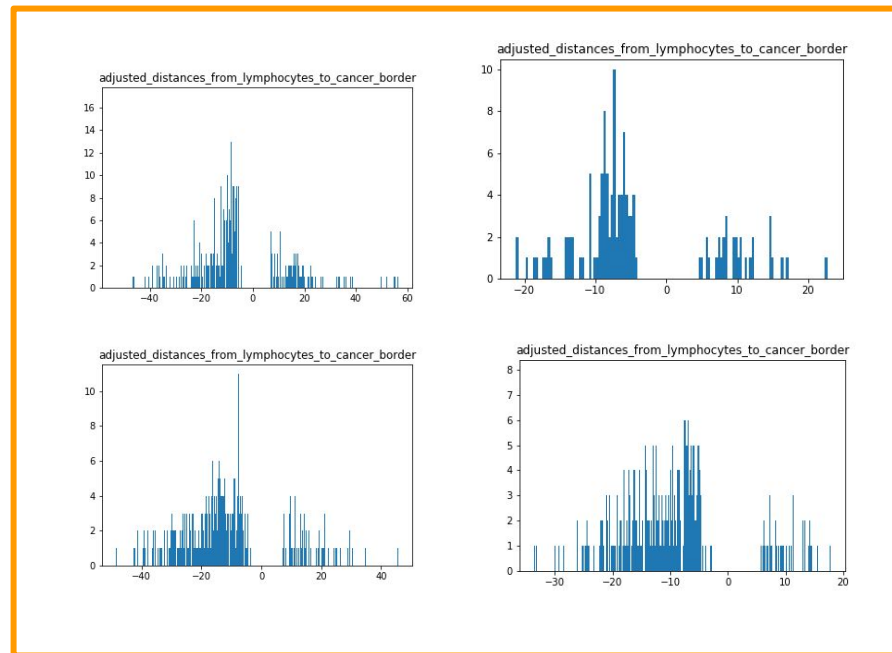
Features	p-values (Log-rank)
Distances to the tumor border	0.0022584655839803943
Minimum distance between 2 types of cells (lymphocyte - cancer)	0.05737550179413786
Area at the interface cancer-stroma	0.34611512869827965
Area at the interfacecancer-lymphocyte	0.11365602326149649
Ratio # lymphocyte / # all the cells	0.019320720457706653
Size of clusters of cells of the same type - cancer	0.19982450948750324
Size of clusters of cells of the same type - lymphocyte	0.034908978593598615
tumeur ratio	0.16864032817841768
lymphocyte density	0.0009978137511020583
Combinaison	10^{-5}

Distances to the tumor border (ajustées) lymphocyte - cancer

p-value : 0.0022584655839803943



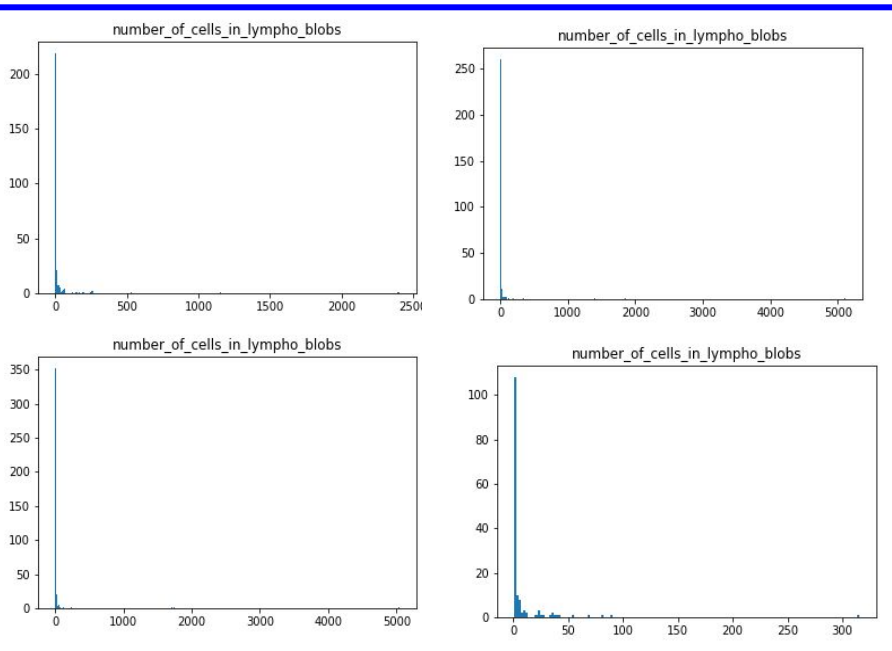
Population 0 top



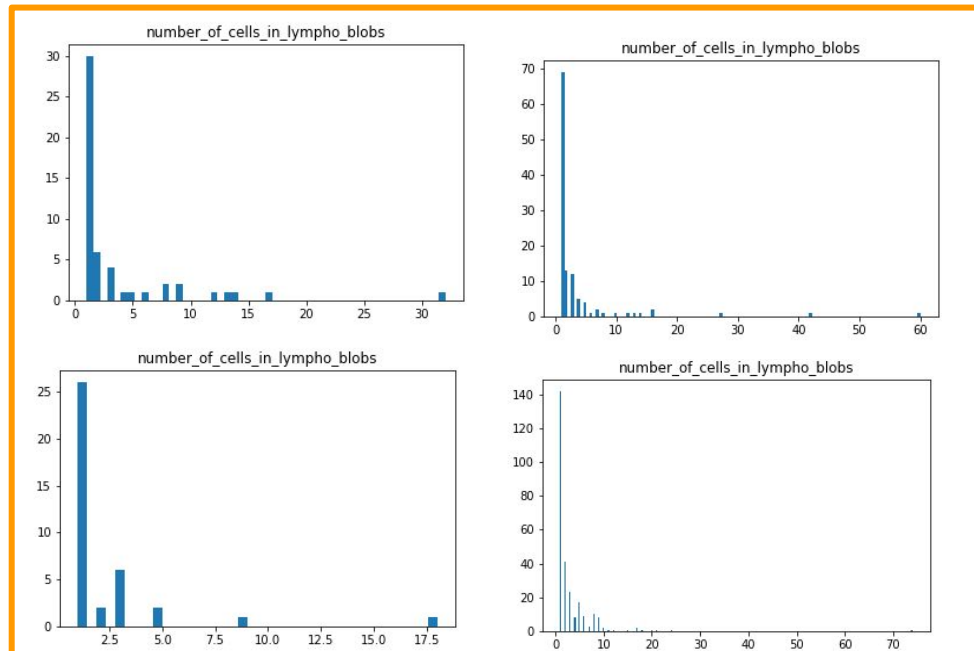
Population 1 bottom

Size of clusters of cells of the same type - lymphocyte

p-value : 0.034908978593598615



Population 0 top

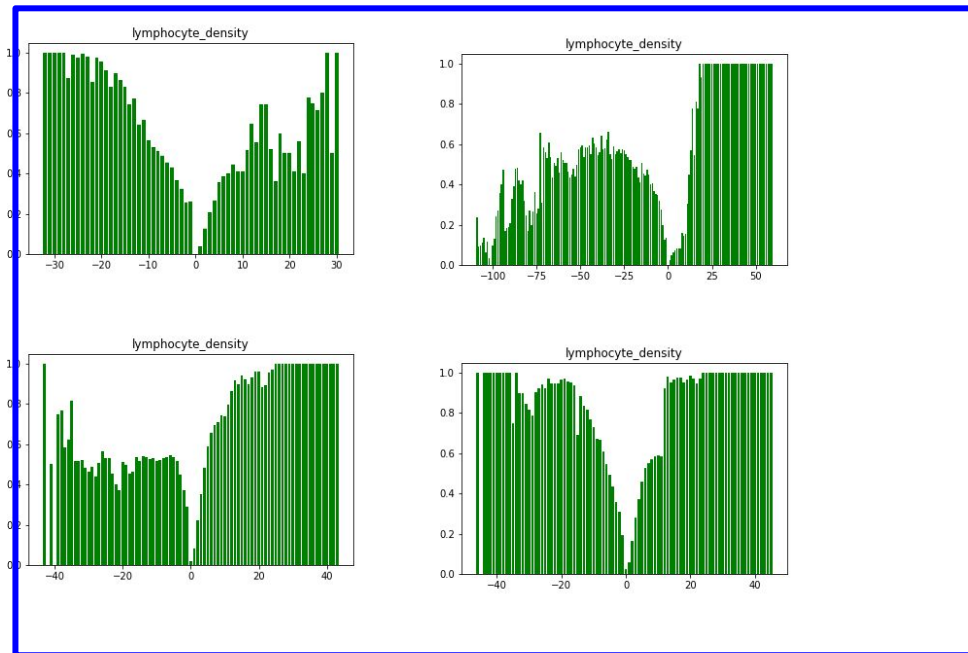


Population 1 bottom

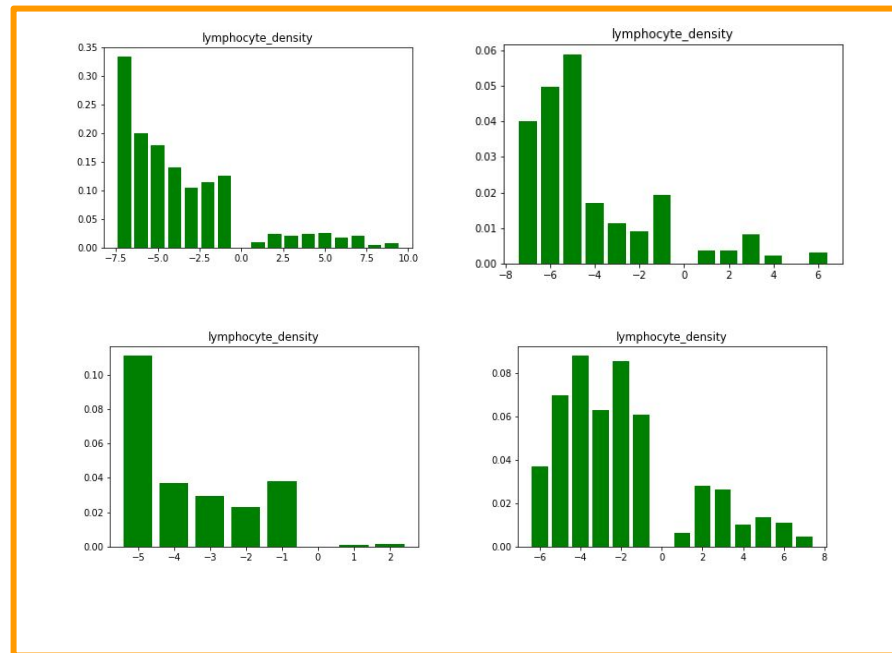
Interprétation : Population 0 has more big size of cluster of lymphocytes → has a better defence against cancer

Lymphocyte density

p-value : 0.0009978137511020583

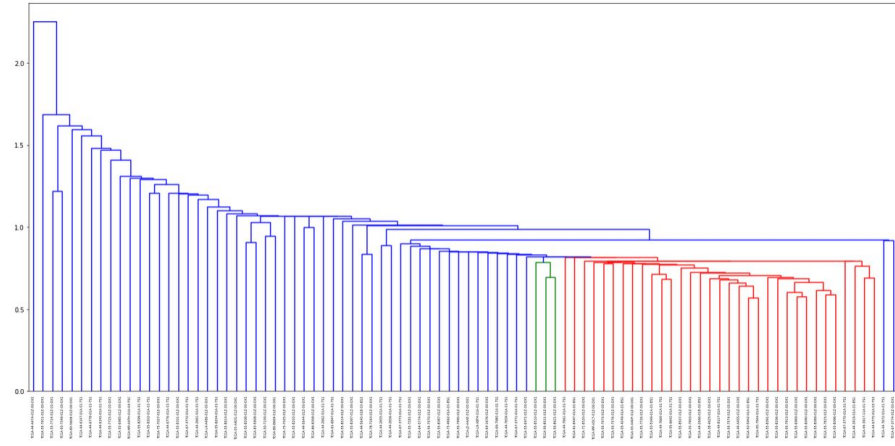
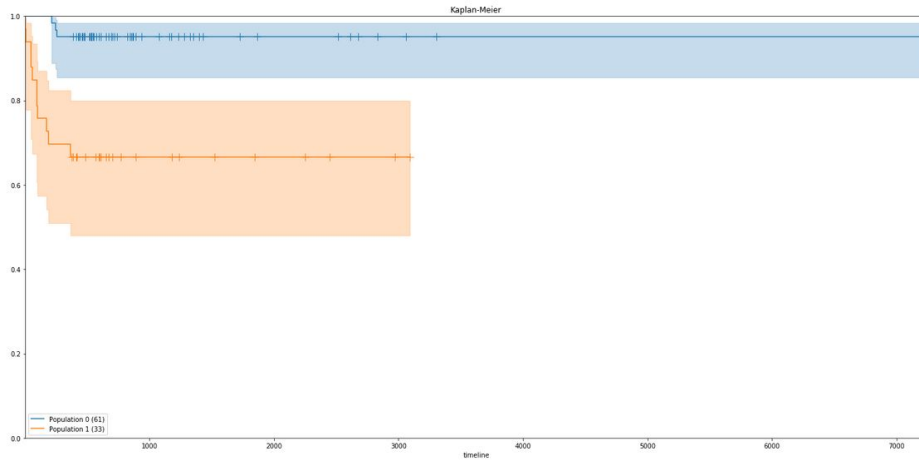


Population 0 top



Population 1 bottom

Best results



p value = $9.97637869 \times 10^{-5} \lll 0.05$